

DLFA - 25 Class Test 1 Soln.

Question 1

$$1. \text{Depth} = 2 \log_2 n = 2 \log_2 16 = 8$$

$\hookrightarrow 2 \text{ layers / level}$

Minimum # of neurons needed to implement a 2 i/p XOR = 3

$$\begin{aligned} \therefore \text{Min}^m \text{ # of neurons} &= (N-1) \times 3 \\ &= 15 \times 3 \\ &= 45 \end{aligned}$$

$$(8 + 4 + 2 + 1) = 15$$

Ans. B.

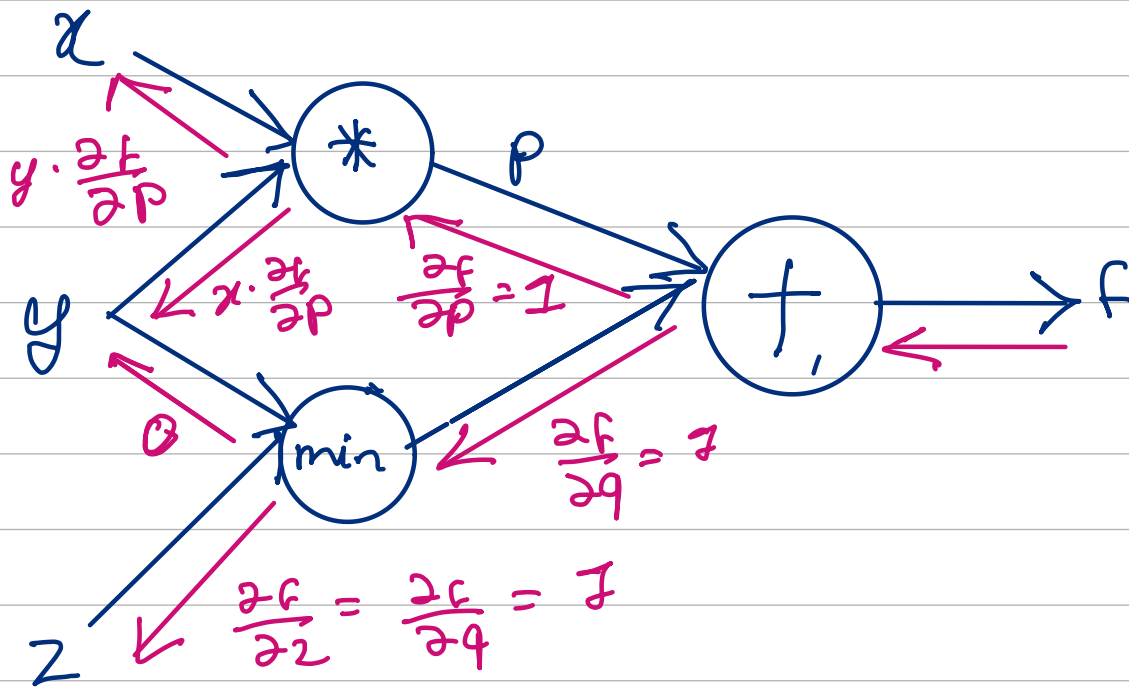
2. Ans. A, C, D

$$3. \text{ # of layers between the i/p of layer 7 and output} = 10 - 7 + 1 = 4$$

$$\therefore \text{ # of gradient flow paths} = 2^4 = 16$$

Ans. C.

Question 2



$$x = 2, y = 0.55, z = -1$$

$$p = xy = 1.1$$

$$q = \min(y, z) = -1$$

$$f = p + q = 0.1$$

$$\frac{\partial f}{\partial p} = 1 \quad \frac{\partial f}{\partial q} = 1$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial q} = 1$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial x}$$

$$= 1 \cdot \frac{\partial (xy)}{\partial x}$$

$$= 1 \cdot y$$

$$= 0.55$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial p} \cdot \frac{\partial p}{\partial y} + \frac{\partial f}{\partial q} \cdot \frac{\partial q}{\partial y}$$

$$= 1 \cdot \frac{\partial (xy)}{\partial y} + 1 \cdot \frac{\partial \min(y, x)}{\partial y}$$

$$= 1 \cdot x + 1 \cdot 0$$

$$= 2$$

Question 3

$$1. \quad w_1 \in \mathbb{R}^{k_1 \times n}$$

$$w_2 \in \mathbb{R}^{k_2 \times k_1}$$

$$w_3 \in \mathbb{R}^{m \times k_2}$$

$$2. \quad b_1 \in \mathbb{R}^{k_1}$$

$$b_2 \in \mathbb{R}^{k_2}$$

$$b_3 \in \mathbb{R}^m$$

$$3. \quad n=5, \quad k_1=8, \quad k_2=4, \quad m=2$$

of learnable parameters

$$(5 \times 8 + 8) + (8 \times 4 + 4) + (4 \times 2 + 2)$$

$$= 94$$

Question 3

$$1 \text{ logits} : z = \begin{bmatrix} 1.2 \\ 0.5 \\ -0.7 \end{bmatrix}$$

$$g(z_i) = \frac{e^{z_i}}{\sum_{k=1}^3 e^{z_k}}$$

$$g(1.2) = \frac{e^{1.2}}{e^{1.2} + e^{0.5} + e^{-0.7}} \\ \approx 0.6075$$

$$g(0.5) = \frac{e^{0.5}}{e^{1.2} + e^{0.5} + e^{-0.7}} \\ \approx 0.3017$$

$$g(-0.7) = \frac{e^{-0.7}}{e^{1.2} + e^{0.5} + e^{-0.7}} \\ \approx 0.0909$$

$$\hat{y} = \begin{bmatrix} 0.6075 \\ 0.3017 \\ 0.0909 \end{bmatrix} \quad (\text{Predicted class likelihood})$$

$$\underline{2.} \quad \frac{\partial \hat{y}_i}{\partial z_i} = e^{z_i} \sum_{k=1}^3 e^{z_k} - e^{z_i} e^{z_i}$$

$$= \frac{e^{z_i}}{\sum_{k=1}^3 e^{z_k}} \left(\frac{\sum_{k=1}^3 e^{z_k} - e^{z_i}}{\sum_{k=1}^3 e^{z_k}} \right)$$

$$= \frac{e^{z_i}}{\sum_{k=1}^3 e^{z_k}} \left(1 - \frac{e^{z_i}}{\sum_{k=1}^3 e^{z_k}} \right)$$

$$= \hat{y}_i (1 - \hat{y}_i)$$

$$\frac{\partial \hat{y}_i}{\partial z_j} = \sum_{k=1}^3 e^{z_k} \frac{\partial}{\partial z_j} (e^{z_i}) - e^{z_i} \frac{\partial}{\partial z_j} \left(\sum_{k=1}^3 e^{z_k} \right)$$

$$\left(\sum_{k=1}^3 e^{z_k} \right)^2$$

$$= \frac{0 - e^{z_i} e^{z_j}}{\left(\sum_{k=1}^3 e^{z_k} \right)^2}$$

$$= - \frac{e^{z_i}}{\sum_{k=1}^3 e^{z_k}} \cdot \frac{e^{z_j}}{\sum_{k=1}^3 e^{z_k}}$$

$$= -\hat{y}_i \hat{y}_j$$

$$\underline{3.} \quad \ell(y, \hat{y}) = -1 \cdot \log(0.6075) \\ = 0.4984$$

$$\underline{4.} \quad \ell(y, \hat{y}) = - \sum_{k=1}^3 y_k \log \hat{y}_k$$

$$= -y_1 \log \hat{y}_1 + 0 + 0$$

$$(\text{As } y = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix})$$

$$\frac{\partial l}{\partial \hat{y}_1} = -\frac{y_1}{\hat{y}_1}, \quad \frac{\partial l}{\partial \hat{y}_2} = 0, \quad \frac{\partial l}{\partial \hat{y}_3} = 0$$

$$\frac{\partial l}{\partial z_1} = \frac{\partial l}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial z_1}$$

$$= -\frac{y_1}{\hat{y}_1} \cdot \hat{y}_1 (1 - \hat{y}_1)$$

$$= -1(1 - \hat{y}_1)$$

$$= \hat{y}_1 - 1$$

$$= 0.6075 - 1$$

$$= -0.3925$$

$$\frac{\partial l}{\partial z_2} = \frac{\partial l}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial z_2}$$

$$= -\frac{y_1}{\hat{y}_1} (-\hat{y}_1 \hat{y}_2)$$

$$= y_1 \hat{y}_2$$

$$= 0.3017$$

$$\frac{\partial l}{\partial z_3} = \frac{\partial l}{\partial \hat{y}_1} \cdot \frac{\partial \hat{y}_1}{\partial z_3}$$

$$= \frac{-y_1}{\cancel{0}} (-\hat{y}_1 y_3)$$

$$= y_1 \hat{y}_3$$

$$= 0.0909$$

$$\therefore \frac{\partial l}{\partial z} = [-0.3925 \quad 0.3017 \quad 0.0909]$$

(numerator layout so row vector)